Lecture 05 (B04): Function Approximation and Loss Design

Universal approximation, robust losses, the curse of dimensionality

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Universal Approximation Theorem

Theorem (Cybenko 1989; Hornik, Stinchcombe & White 1989)

Let $g : \mathbb{R} \to \mathbb{R}$ be a bounded, non-constant, continuous activation function. Then for any continuous function $f : \mathcal{K} \to \mathbb{R}$ on a compact set $\mathcal{K} \subset \mathbb{R}^d$ and any $\varepsilon > 0$, there exists a single-hidden-layer network

$$F(\boldsymbol{x}) = \sum_{j=1}^{n_1} v_j \, g(\boldsymbol{w}_j^\top \boldsymbol{x} + b_j)$$

such that $\sup_{\boldsymbol{x} \in \mathcal{K}} |F(\boldsymbol{x}) - f(\boldsymbol{x})| < \varepsilon$.

Caveats:

- The required width n_1 may be **exponentially large** in d.
- Existence $\neq$ efficient learnability.
- Practical success of deep learning suggests that **depth** provides a much more efficient parameterization than width alone.

Part III

Training: Loss, Gradients, and Optimization

Loss Functions

The loss function $J(\boldsymbol{\theta})$ measures how far predictions $\hat{\boldsymbol{y}}$ are from targets $\boldsymbol{y}$.

Regression: Mean Squared Error

$$J(\boldsymbol{\theta}) = \frac{1}{2n} \sum_{i=1}^{n} \left(y^{(i)} - f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta})\right)^2$$

Penalizes large deviations quadratically.

Classification: Cross-Entropy

$$J = -\frac{1}{n} \sum_{i=1}^{n} \left[y^{(i)} \log \hat{y}^{(i)} + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right]$$

For binary labels $y \in \{0, 1\}$.

Multiclass (K classes): use softmax output $\hat{y}_k = e^{z_k} \big/ \sum_j e^{z_j}$ and categorical cross-entropy: $J = -\frac{1}{n} \sum_i \sum_k y_k^{(i)} \log \hat{y}_k^{(i)}$.

Beyond MSE: Robust and Asymmetric Losses

Let $r = y - \hat{y}$ denote the residual.

MAE (L1): robust to outliers (constant gradient)

$$L_{\text{MAE}} = \frac{1}{n}\sum_{i=1}^{n} \left|r^{(i)}\right|$$

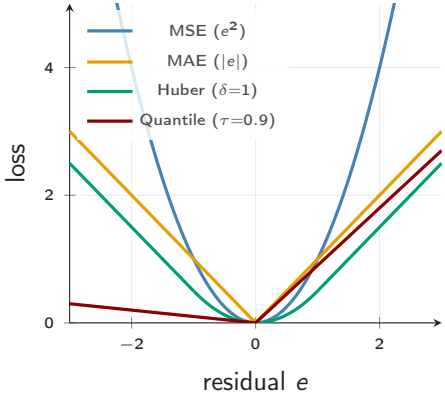
Huber (δ): smooth near zero, linear in the tails

$$L_\delta(r) = \begin{cases} \frac{1}{2}r^2 & |r| \le \delta \\ \delta\left(|r| - \frac{\delta}{2}\right) & |r| > \delta \end{cases}$$

Quantile / Pinball ($\tau \in (0,1)$):

$$L_\tau(r) = \begin{cases} \tau\,|r| & r \ge 0 \ \text{(under-predict)} \\ (1-\tau)\,|r| & r < 0 \ \text{(over-predict)} \end{cases}$$

$\tau > 0.5$: penalises under-prediction more $\Rightarrow$ predicts *high* quantile. Directly gives VaR at level τ.



Choosing a Loss Function

Loss	Outliers	Smoothness	Symmetry	Use case
MSE	High	Smooth	Symmetric	Default regression
MAE	Low	Non-smooth at 0	Symmetric	Median regression
Huber	Low	Smooth	Symmetric	Robust regression
Quantile	Low	Non-smooth at 0	**Asymmetric**	Risk, VaR, tails

Why this matters in economics and finance

Tail risks are often more important than average performance. **Quantile loss** lets the model focus on getting worst-case predictions right, directly producing Value-at-Risk (VaR) or Expected Shortfall estimates. See **Notebook 07** for a hands-on comparison.

References: Huber (1964), Koenker & Bassett (1978).

Exercise: Function Approximation with Neural Networks

Task: approximate the Genz (1987) test functions with neural networks.

$$f_1(\boldsymbol{x}) = \cos\Big(2\pi w_1 + \textstyle\sum_{i=1}^d c_i x_i\Big) \qquad \text{(oscillatory)}$$

$$f_2(\boldsymbol{x}) = \textstyle\prod_{i=1}^d \left(c_i^{-2} + (x_i - w_i)^2\right)^{-1} \qquad \text{(product peak)}$$

$$f_3(\boldsymbol{x}) = \Big(1 + \textstyle\sum_{i=1}^d c_i x_i\Big)^{-(d+1)} \qquad \text{(corner peak)}$$

$$f_4(\boldsymbol{x}) = \exp\Big(-\textstyle\sum_{i=1}^d c_i^2 (x_i - w_i)^2\Big) \qquad \text{(Gaussian)}$$

$$f_5(\boldsymbol{x}) = \exp\Big(-\textstyle\sum_{i=1}^d c_i |x_i - w_i|\Big) \qquad \text{(continuous)}$$

$$f_6(\boldsymbol{x}) = \begin{cases} 0 & \text{if } x_1 > w_1 \text{ or } x_2 > w_2 \\ \exp\big(\sum_{i=1}^d c_i x_i\big) & \text{otherwise} \end{cases} \qquad \text{(discontinuous)}$$

Pick 3 functions. Train NNs on 10, 50, 100, 500 random points from $[0,1]^d$. Compute average and max error on 1000 test points. Repeat for $d \in \{2, 5, 10\}$.

Questions?

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https://sischei.github.io

Materials: https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models